Learned Deferral for Auditable Bounded Decision Control

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Abstract

Production AI systems often operate with incomplete, conflicting, or insufficient evidence. Forced classifiers collapse such cases into action labels, while generative systems can produce outputs that are difficult to interpret as auditable execution decisions. This paper presents a bounded decision-control model, EvaluatorDPT, that predicts YES, NO, or TBD, where TBD is learned as a deferral outcome rather than added only as a post-hoc confidence rule. The model uses a transformer encoder with a primary bounded-decision head and structured auxiliary channels for values and emotions/sentiments. The paper frames this task as bounded decision control: a model must distinguish action, rejection, and deferral while preserving enough evidence for operating-policy review. For the evaluated S12B checkpoint, we report decision performance on DS-L validation and test splits; auxiliary emotion metrics are omitted because the emotion head is masked in this lineage. On the full DS-L test split (n=44,597), S12B achieves Accuracy = 0.8260 and Macro F1 = 0.8252, with per-class F1 of 0.8314 (YES), 0.8486 (NO), and 0.7956 (TBD). The evaluation record also includes calibration evidence (ECE = 0.0338 on validation), threshold-sweep artifacts, multi-seed stability checks, confusion matrices, and reproducibility commands. The main contribution is a bounded decision formulation in which deferral is learned, operating thresholds remain inspectable, and evaluation artifacts support external review.

1 Introduction

Many deployed systems now connect language inputs to downstream actions. In such settings, classification accuracy alone is not the central question; the system must also decide whether to act, block, or defer. Inputs may be incomplete, contradictory, or ambiguous. A model that must always choose YES or NO can conceal uncertainty in cases where an operator would prefer review.

The issue appears when language models or automated agents are connected to business workflows, policy enforcement, moderation, approvals, safety checks, or other action-taking systems. In these environments, a model output is not merely a label. It becomes a control signal. A false YES may trigger an unsafe or unauthorized action; a false NO may block a valid action; and a failure to defer may hide uncertainty from the surrounding workflow. A useful decision model therefore needs an explicit representation of uncertainty, rather than a confidence score attached to a forced label.

The model is designed for this setting. It is not a general-purpose language generator or a replacement for domain-specific review. It is a bounded classifier that converts a context signal into a constrained decision with confidence and supporting evaluation evidence. Downstream policies can inspect the decision distribution, choose operating thresholds, route deferred cases, and preserve a review trail.

The proposed formulation models decision control as a bounded three-way output space:

$$\mathcal{Y}_d = \{\text{YES}, \text{NO}, \text{TBD}\}.$$

The TBD outcome is not merely a post-hoc rejection threshold. It is a learned class that allows the model to internalize deferral boundaries during training. Thresholding remains useful at deployment time, but it is applied on top of a model that already represents ambiguity as part of the label space.

The contribution of this work is **learned deferral as an auditable decision-control formulation**. The model does not generate free-form decisions. It emits bounded labels, confidence scores, and structured auxiliary signals, while the evaluation package provides the artifacts needed to inspect behavior: calibration, threshold sweeps, multi-seed stability, confusion matrices, and reproducibility materials.

This framing is intentionally narrower than many general language-model evaluations. The paper does not argue that a single model can resolve all downstream governance questions. Instead, it studies a smaller but operationally important problem: how to expose an explicit, learned deferral decision in a form that can be evaluated, thresholded, logged, and reviewed. The design goal is not maximal automation. It is controlled routing under ambiguity.

Contributions.

- We formulate decision routing as bounded control over YES, NO, and learned TBD deferral.
- We present a transformer-based multi-head architecture that separates primary decision prediction from structured auxiliary context signals.
- We describe targeted boundary refinement for hard ambiguity cases, including negation, contradiction, lexical-overlap traps, and modal/hedged statements.
- We evaluate the S12B checkpoint on full validation and test splits and report per-class behavior, calibration, threshold-sweep evidence, and stability checks.
- We organize evaluation results, calibration data, threshold sweeps, and reproducibility materials into a coherent review structure.

2 Problem Formulation

This work studies a decision-control setting rather than a standard text-classification setting. A model receives a context signal x and must output a decision that can be consumed by a downstream workflow. The decision may authorize an action, block an action, or defer the action for additional review. The output is therefore not only a semantic label; it is an interface between language understanding and operational policy.

We define the bounded decision space as

$$\mathcal{Y}_d = \{\text{YES}, \text{NO}, \text{TBD}\}.$$

YES denotes sufficient support for action, NO denotes sufficient support for rejection or blocking, and TBD denotes insufficient certainty for either direct action or direct rejection. In deployment, TBD may route to human review, a policy engine, evidence collection, or another conservative fallback.

This formulation differs from binary classification with a reject threshold in two ways. First, deferral is represented in the supervised label space, so ambiguous examples can influence the learned representation. Second, thresholding remains available as an operating-policy layer over a three-way decision distribution. The model can learn the semantic structure of deferral, while deployment owners can still select thresholds that reflect local risk tolerance.

The practical requirement is not only high aggregate performance. A decision-control model should also expose the evidence needed to review its behavior: per-class outcomes, confusion patterns, calibration, threshold tradeoffs, stability under repeated evaluation, and artifact identity. These requirements motivate the certification-oriented release structure described later in the paper.

2.1 Learned Deferral and Reject-Threshold Abstention

Let $p_\theta(y \mid x)$ be the model distribution over $\mathcal{Y}_d$. In a standard reject-threshold design, a binary classifier first estimates $p_\theta(y \in \{\mathrm{YES}, \mathrm{NO}\} \mid x)$ and then abstains when confidence, entropy, or margin fails a threshold. Deferral is therefore an operating rule applied after the classifier has learned a binary decision boundary.

In this work, deferral is part of the supervised decision space. The training objective includes examples labeled TBD, so the model can assign probability mass directly to an insufficient-evidence outcome:

$$\hat{y} = \arg \max_{y \in \{\text{YES,NO,TBD}\}} p_\theta(y \mid x).$$

A deployment threshold τ_y may still be applied:

$$g(x) = \begin{cases} \arg\max_y p_\theta(y \mid x), & \exists y : p_\theta(y \mid x) \geq \tau_y, \\ \text{TBD}, & \text{otherwise.} \end{cases}$$

The distinction is not that thresholds are avoided. The distinction is that thresholds operate over a distribution that already contains a learned deferral class. The experiments below therefore evaluate two separate questions: the value of a learned three-way decision space at full coverage, and the operating tradeoff obtained by applying reject thresholds to model confidence.

3 What Is Novel in This Work

The novelty is not the use of a transformer encoder by itself. The contribution lies in the way the model and evaluation protocol are organized around bounded decision control.

Learned deferral. TBD is treated as a supervised outcome, not only as a threshold applied after a YES/NO prediction. This allows ambiguous and insufficient-evidence examples to participate in training. The model can therefore learn patterns that correspond to deferral rather than treating all low-confidence regions as equivalent.

Separation of decision and operating policy. The model emits a bounded decision distribution, while deployment thresholds remain external and inspectable. This separates the representation of uncertainty from the choice of how conservatively the system should act in a particular environment.

Release-level auditability. The evaluation package includes calibration data, threshold-sweep artifacts, multi-seed stability checks, confusion matrices, reproducibility commands, and model/data pointers. These artifacts make the evaluated behavior reviewable. This is a different standard from publishing only model weights and a top-line score.

Boundary-focused refinement. The development process identifies deferral boundaries as the key failure region and uses targeted boundary examples rather than undifferentiated data growth. This is especially relevant for negation, contradiction, lexical-overlap traps, long-text stress cases, and hedged statements.

Together, these choices define the scope of the paper. The model is presented as a bounded formulation for decision routing, not as a new language-model backbone. The empirical question is whether a learned deferral class, evaluated with classwise and calibration evidence, can support controlled decision workflows.

3.1 Design Principles

The work is guided by six design principles for auditable decision routing:

1. Uncertainty should be explicit, not hidden inside a forced YES/NO decision.
2. Deferral should be represented as a first-class outcome when training data supports it.
3. Operating thresholds should remain external to the checkpoint and be recorded as policy.
4. Confidence should be accompanied by calibration evidence.
5. Evaluation should report classwise behavior, not only aggregate accuracy.
6. The evaluated artifact, split, threshold policy, and commands should be reconstructable.

4 Related Work

The proposed model is related to selective classification and reject-option models, which abstain from prediction under uncertainty [5, 7, 6]. In many such systems, abstention is applied after prediction using confidence thresholds. Here, the model instead learns a deferral label directly while still preserving threshold sweeps for operating-point selection. A learned TBD class allows ambiguous examples to shape the representation during training, while thresholding remains a deployment policy rather than the only source of abstention.

The architecture builds on transformer language models such as BERT [4] and multi-task learning [2]. Here, auxiliary outputs are treated as structured context signals available at inference time rather than only as training regularizers.

The auxiliary channels also differ from conventional sentiment or lexical classification tasks. In this work, value and emotion/sentiment signals are interpreted relative to the decision event. They are intended to encode decision-relevant semantic context rather than surface affect alone. This distinction is important because a sentence can contain positive or negative language without providing sufficient support for a YES or NO decision.

Neural confidence scores can be miscalibrated [9], so the evaluation pairs decision outputs with calibration evidence and threshold-sweep artifacts. This is relevant for governance because the operating policy should be inspectable rather than implicit.

The paper is also related to model cards and governance-oriented reporting practices. Such artifacts often document intended use, limitations, and evaluation results after training. In this work, these materials are treated as part of the evaluation record: the model is accompanied by structured metrics, calibration data, threshold sweeps, stability checks, and reproducibility commands.

5 System Overview

The model receives a context signal and returns:

- a bounded decision: YES, NO, or TBD;
- decision confidence derived from the primary decision head;
- structured auxiliary signals for values and emotions/sentiments, where enabled by the lineage;
- evidence artifacts that support threshold selection and audit.

The intended deployment pattern is conservative. YES and NO represent bounded action signals. TBD represents a case for human review, a policy engine, additional evidence collection, or another fallback. Because the outputs are bounded, downstream systems can log the decision distribution and threshold policy without parsing generated text.

The system boundary is therefore clear. The model does not execute the downstream action, resolve policy conflicts, or replace review. It provides a decision-control signal that can be consumed by another system. A deployment can use the raw argmax decision, enforce a minimum confidence margin, route all TBD cases to review, or apply a more conservative threshold policy derived from the threshold-sweep artifacts. This keeps the model output separate from the operational decision rule.

5.1 Operational Interface

In an operational setting, each prediction produces a record containing the input identifier, model version, checkpoint identity, decision probabilities, selected operating threshold, final routed label, and any auxiliary channels enabled for the lineage. This structure is important because reviewability depends on reconstructing not only the model score but also the policy applied at the time of decision.

The interface also supports conservative fallback. A downstream system can treat low-confidence YES or NO predictions as TBD without retraining the model. Conversely, if a use case tolerates more automation, it can lower the deferral threshold while preserving the same model artifact. This is the reason threshold selection is treated as release evidence rather than as an implementation detail.

6 Methodology

6.1 Bounded Decision Formulation

Let x be a decision-event input. The model predicts a primary decision distribution over $\mathcal{Y}_d = \{\mathrm{YES}, \mathrm{NO}, \mathrm{TBD}\}$. The TBD label represents explicit deferral under insufficient semantic certainty. This differs from binary classification followed by confidence filtering because the model learns examples of deferral as part of the supervised task.

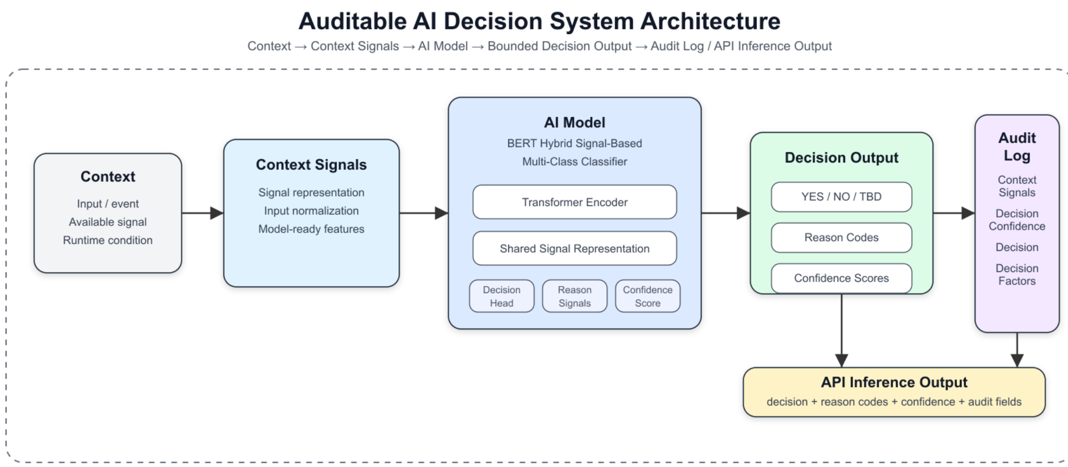

Figure 1: System architecture: tokenized decision event to shared transformer encoder, bounded decision output, confidence, and structured auxiliary channels.

Deployment policies may still choose operating thresholds over the predicted probabilities. In this formulation, thresholding is an operating policy layered over a bounded decision model, not the sole mechanism by which uncertainty is represented.

This formulation gives the system two mechanisms for risk control. The first is learned: ambiguous or insufficient contexts can be labeled TBD during training. The second is operational: deployment owners can choose thresholds over the learned distribution to match risk tolerance. A high-assurance setting may route more cases to TBD, while a lower-risk setting may accept a smaller margin for YES or NO. The model separates the representation of uncertainty from the policy decision about how much uncertainty is acceptable.

6.2 Architecture

The architecture uses a BERT-base encoder [4] with a maximum sequence length of 128 tokens. The shared encoder feeds task-specific heads:

- a primary three-class decision head;
- a 10-label value head;
- a 28-label emotion/sentiment head.

For S12B, the decision head is the validated output reported in this paper. The emotion head is masked in the DS-L lineage, so emotion metrics are intentionally not claimed for this checkpoint. The auxiliary-head architecture remains part of the model design, but this paper scopes empirical claims to the validated S12B decision behavior and evaluation evidence.

6.3 Training Configuration

S12B starts from the S12 DS-L checkpoint and applies one boundary-pack sharpening epoch. The training configuration uses BERT-base-uncased, maximum sequence length 128, batch size 32, evaluation batch size 64, cosine scheduling with warmup ratio 0.05, label smoothing 0.05, and loss weights 1.0, 0.3, and 0.3 for the decision, value, and emotion/sentiment heads respectively. Encoder layers 9–11 are unfrozen with layer-wise learning rates 1.0×10^{-6}, 7.5×10^{-7}, and 5.0×10^{-7}. The global learning rate is 5.0×10^{-6} and the recorded seed is 42. The emotion head is masked in the DS-L lineage during evaluation, so no emotion-head metric is reported for S12B.

6.4 Data and Source Scope

The DS-L lineage is a curated decision-event corpus built from public NLP sources and curated decision examples. The corpus is organized around bounded decision labels and auxiliary context signals. The repository does not redistribute raw training data. Instead, it provides dataset documentation, processing notes, and artifact pointers sufficient for review of the evaluated checkpoint.

The dataset should be interpreted as a research and evaluation corpus for bounded decision behavior, not as a universal production benchmark. The paper evaluates learned deferral under the reported distribution; it does not claim universal generalization to all domains.

The lineage includes three broad kinds of material. First, it uses natural-language inference style examples to expose entailment, contradiction, and neutral/insufficient-evidence patterns. Second, it includes curated decision examples that map language contexts into the bounded YES/NO/TBD control space. Third, it adds targeted boundary material for cases that were difficult in earlier diagnostics, including number changes, role changes, negation, high-overlap contradiction, low-overlap support, long-context stress, and modal or hedged language. These categories are described at a level suitable for publication; reuse of upstream datasets must follow the licenses of the original sources.

Split	YES	NO	TBD
Train	137,168	149,071	132,486
Validation	14,962	15,534	13,908
Test	14,883	15,650	14,064

Table 1: Decision-class support for the evaluated DS-L validation and test splits.

DS-L contains 418,725 training examples, 44,404 validation examples, and 44,597 test examples. The validation and test splits are fixed from the DS-J lineage and are reused unchanged for the S12/S12B evaluations. The S12B boundary pack contains 323 training examples mined from S12 diagnostics; it is used only for one sharpening epoch and is less than 0.1% of the DS-L training volume.

6.5 Annotation and Construction Transparency

The decision labels are derived from a natural-language inference formulation. Entailment maps to YES, contradiction maps to NO, and neutral or insufficient evidence maps to TBD. No new project-specific inter-annotator agreement study was conducted for S12B. The project inherits the annotation assumptions of the source NLI datasets, including reported agreement for SNLI and MultiNLI, and applies documented correction passes where diagnostics identified systematic label errors.

The main dataset construction steps are also recorded. DS-B contained approximately 814K examples after initial cleaning. DS-E reduced the corpus to the NLI-compatible subset, producing the largest quality gain in the lineage. DS-J became the canonical fixed split baseline. DS-L expanded the DS-J training split to 418,725 rows by adding targeted hard cases and capped pattern boosts while keeping validation and test splits unchanged. The exact per-source breakdown inside the 323-row boundary pack was not preserved in the final CSV; this is recorded as a dataset-transparency limitation rather than inferred after the fact.

6.6 Label Semantics

The labels are defined by decision sufficiency rather than by surface sentiment. YES means the input provides enough support for the affirmative action or decision. NO means the input provides enough support to reject or block. TBD means the input lacks sufficient support for either direct action or direct rejection under the modeled policy. This distinction matters because a positive-sounding sentence may still be TBD if it does not provide enough evidence, and a negative-sounding sentence may be YES if it supports an affirmative control decision in context.

The auxiliary value and emotion/sentiment channels are designed to represent structured context, not to replace the primary decision label. Their intended semantics are decision-relative: the labels describe values and affective context as they bear on the decision event, not merely the linguistic tone of the text. For example, lexical positivity does not imply a YES label, and lexical negativity does not imply a NO label. The relevant question is whether the text provides decision-sufficient support, rejection, or uncertainty.

In S12B, the empirical claims are restricted to the decision head because the emotion head is masked in the DS-L lineage. This avoids overstating auxiliary performance and keeps the paper aligned with the validated artifact. Future auxiliary-head evaluation should therefore test decision-semantic alignment rather than only agreement with surface sentiment or keyword-based labels.

6.7 Boundary Refinement

A recurring challenge in bounded decision systems is the boundary between action and deferral. S12B uses DS-L with a targeted boundary pack focused on hard semantic regions such as contradiction, negation, low lexical overlap entailment, high lexical overlap contradiction, long-text stress cases, and hedge/modal ambiguity. The goal is to improve boundary behavior using targeted examples rather than relying only on broad dataset scale.

This design reflects an empirical lesson from the development lineage: additional data is most useful when it targets the failing boundary. The boundary pack focuses on cases where surface overlap can be misleading, where negation changes the decision, where numbers or roles alter entailment, or where a statement is too hedged to support a direct YES or NO. These cases matter for learned deferral because the model must distinguish confident action from insufficient evidence.

7 Evaluation Setup

The evaluated checkpoint is `exp_t90_S12B_boundarypack_ep1_fromS12ep3`. Results are reported for DS-L validation and test splits:

- validation: n = 44,404;
- test: n = 44,597.

The evaluation materials include full-split metrics, confusion matrices, calibration data, threshold sweeps, and reproducibility commands.

The primary metric is Macro F1 because the decision labels must be evaluated across all three outcomes rather than dominated by the largest class. Accuracy is reported as a secondary metric. Per-class precision, recall, and F1 are reported to expose tradeoffs among action, rejection, and deferral.

7.1 Evaluation Questions

The evaluation is organized around four questions:

1. Does the model maintain balanced decision performance across YES, NO, and TBD?
2. Does the deferral class behave as a learned outcome rather than as an afterthought to binary classification?
3. Are the confidence scores sufficiently calibrated to support threshold review?
4. Can an external reviewer identify the evaluated checkpoint, data split, metrics, and commands used to reproduce the reported evidence?

These questions are deliberately practical. A model used as a decision-control layer must be evaluated not only by aggregate score but also by the quality of its failure modes and the traceability of its evaluation artifacts.

7.2 Metrics and Their Roles

Macro F1 measures whether performance is balanced across the bounded decision classes. Per-class F1 identifies which operational boundary is weakest. Precision and recall show whether each class is over-selected or under-selected. The confusion matrix exposes where errors move: for example, whether true TBD cases are being turned into action labels. Calibration error measures whether predicted confidence can be used in threshold policies. Threshold sweeps expose the coverage and deferral tradeoffs available to deployment owners.

The paper reports these metrics together because no single number captures the quality of a deferral-aware system. A higher aggregate F1 would be less useful if it concealed premature action on ambiguous inputs. Conversely, a conservative threshold could increase deferral while reducing automation. The evaluation artifacts are intended to make these tradeoffs explicit.

8 Results

8.1 Decision Performance

Split	Accuracy	Macro F1
Validation (n=44,404)	0.8224	0.8213
Test (n=44,597)	0.8260	0.8252

The validation and test results are close, which supports the stability of the reported behavior under the fixed DS-L split. The test Macro F1 of 0.8252 should be interpreted together with the per-class table below because the operational meaning of errors differs across YES, NO, and TBD.

8.2 Per-Class Test Performance

Class	Precision	Recall	F1	Support
YES	0.8205	0.8425	0.8314	14,883
NO	0.8598	0.8376	0.8486	15,650
TBD	0.7955	0.7958	0.7956	14,064

NO has the highest class F1 in the S12B test result (F1 = 0.8486), indicating that the model learned a useful rejection boundary. TBD remains the hardest class (F1 = 0.7956), which is expected in a defer-aware system because the deferral region is defined by ambiguity rather than by a single semantic pattern. For this reason, the evaluation includes threshold sweeps and calibration evidence rather than relying only on a single argmax policy.

8.3 Abstention and Coverage Baselines

The full-coverage comparison separates learned deferral from forced binary classification. When the same outputs are collapsed into a binary YES/NO decision with no abstention, Macro F1 drops to 0.4945 and TBD F1 is zero because the model can no longer represent deferral. The learned three-way model at full coverage achieves Macro F1 = 0.8252.

Reject-threshold baselines answer a different question: what happens if a system abstains on a fraction of cases and evaluates only the retained subset? At 90% retained coverage, confidence, entropy, and margin rejection produce similar Macro F1 values near 0.863. These numbers are not directly comparable to the full-coverage learned-TBD score because 10% of cases are withheld from the retained set. They do show that thresholding remains a useful operating-policy layer on top of the learned distribution.

Method	Coverage	Macro F1	YES F1	TBD F1
Learned TBD, argmax	1.00	0.8252	0.8314	0.7956
Binary YES/NO, no abstention	1.00	0.4945	0.7482	0.0000
Confidence reject	0.90	0.8631	0.8667	0.8358
Entropy reject	0.90	0.8624	0.8681	0.8329
Margin reject	0.90	0.8625	0.8662	0.8358

8.4 Validation Confusion Matrix

	Pred YES	Pred NO	Pred TBD
True YES	12,496	999	1,467
True NO	1,035	13,066	1,433
True TBD	1,793	1,160	10,955

The validation confusion matrix shows that the largest residual issue is confusion between true TBD and direct action labels, especially TBD predicted as YES. This is the main operational boundary for future improvement: reducing premature action while preserving useful YES/NO coverage.

8.5 Calibration and Stability

The S12B evaluation reports validation ECE = 0.0338. This value indicates that the confidence distribution is suitable for inspection, but it should not be treated as a universal guarantee.

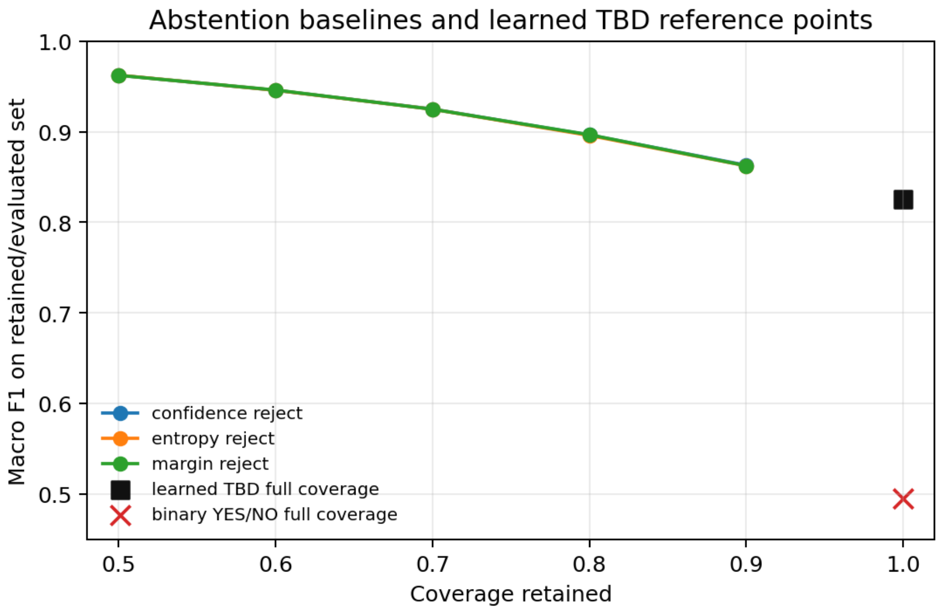

Figure 2: Coverage-sensitive abstention baselines. Reject methods improve retained-set Macro F1 as coverage decreases; learned TBD and forced binary classification are shown as full-coverage reference points.

Calibration is evaluated on the DS-L validation split and should be rechecked after domain transfer or threshold changes.

The evaluation also includes a multi-seed stability check over validation seeds 42, 0, and 7, which produced identical Macro F1 values in the published validation procedure (std = 0.0). This is not a claim that all future training runs will be identical. It shows that the evaluated inference and scoring procedure is deterministic under the recorded configuration.

8.6 S12 to S12B Boundary-Pack Effect

S12B should not be interpreted as a material raw-F1 improvement over S12. On the fixed test split, Macro F1 is essentially unchanged: S12 = 0.8254 and S12B = 0.8252. The measurable effect of the 323-row boundary pack is instead calibration and high-confidence error reduction. In the paired S12/S12B comparison script, validation ECE improves from 0.0461 to 0.0413, test ECE improves from 0.0426 to 0.0411, and high-confidence error rate at threshold 0.85 drops from 0.0558 to 0.0485 on the test split. These paired-comparison ECE values are reported separately from the certification calibration report, which records validation ECE = 0.0338 under its stored binning procedure.

Metric	S12 val	S12B val	S12 test	S12B test
Macro F1	0.8211	0.8213	0.8254	0.8252
ECE	0.0461	0.0413	0.0426	0.0411
High-conf. error @0.85	0.0574	0.0499	0.0558	0.0485
TBD rate	0.3183	0.3120	0.3221	0.3155

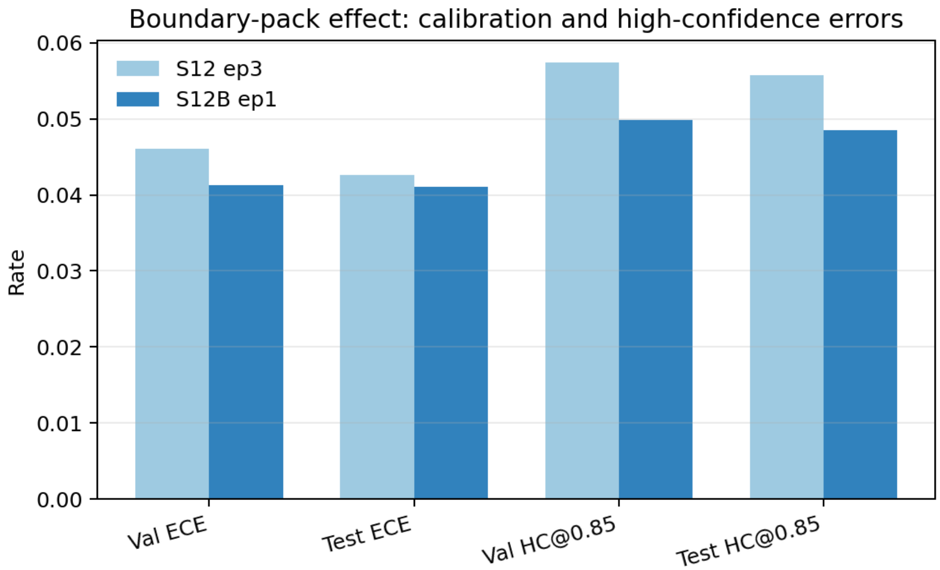

Figure 3: S12 to S12B comparison. The boundary-pack sharpening pass leaves Macro F1 nearly flat but reduces calibration error and high-confidence error rates.

8.7 Interpretation of the Remaining Error Mode

The most important residual error is not a simple lack of class balance. YES, NO, and TBD have comparable support in the test split, and the model performs strongly on NO. The remaining difficulty is semantic: a true TBD case may resemble a YES or NO case because it shares terms, roles, or surface structure while lacking decisive evidence. This is why the boundary pack emphasizes contradiction, negation, role and number swaps, high-overlap non-entailment, and hedged or modal statements.

From an operational perspective, the highest-risk error is a true TBD routed as YES or NO. Such cases reduce the value of deferral because they turn ambiguous inputs into direct actions. The threshold-sweep artifacts are therefore not optional supplementary material; they are part of how a deployment owner decides how much automation to allow.

The qualitative audit on the full test split records 7,758 errors out of 44,597 examples (17.4%). The largest error categories are TBD routed to YES (1,729), NO routed to TBD (1,528), YES routed to TBD (1,350), and TBD routed to NO (1,143). Hard binary confusions are smaller but operationally important: YES routed to NO occurs 994 times and NO routed to YES occurs 1,014 times. The audit examples indicate that the residual errors concentrate around lexical overlap, missing evidence, negation/polarity ambiguity, and cases where a plausible inference goes beyond the premise.

Error category	Count	Mean confidence	Mean margin
TBD→YES	1,729	0.7391	0.5386
TBD→NO	1,143	0.6698	0.4125
YES→TBD	1,350	0.7207	0.5063
NO→TBD	1,528	0.7376	0.5355
YES→NO	994	0.7059	0.4887
NO→YES	1,014	0.7165	0.5107

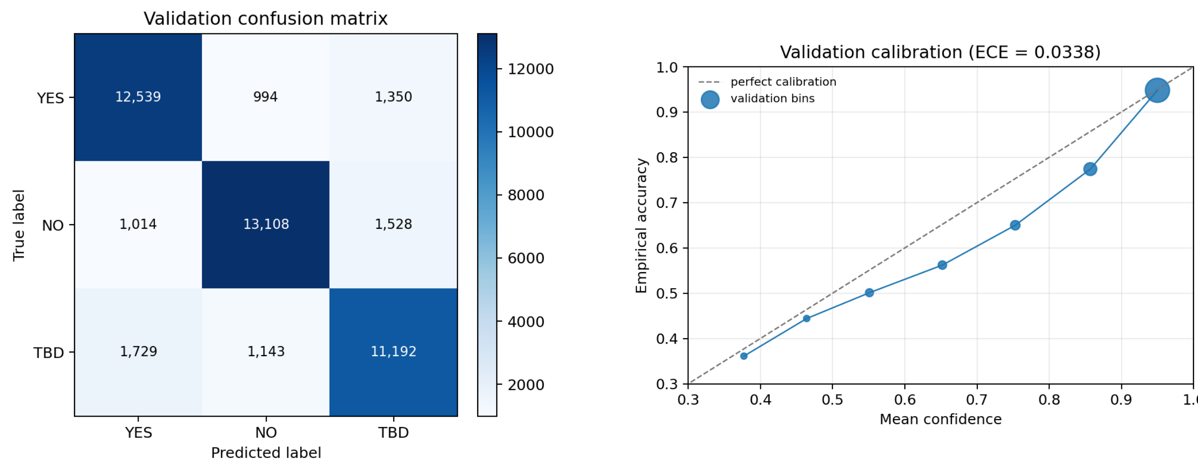

Figure 4: Evaluation artifacts derived from the S12B certification record. Left: validation confusion matrix. Right: validation calibration curve with ECE = 0.0338.

9 Audit-Oriented Evaluation Evidence

The checkpoint is documented with audit-oriented artifacts in addition to evaluation scores. For S12B, the public evidence directory is:

```
certification/runs/S12B_20260526/
```

The directory includes:

- evaluation and calibration reports;
- calibration data with validation ECE = 0.0338;
- threshold-sweep artifacts for TBD-fallback operating policies;
- multi-seed stability evidence, with validation seeds 42, 0, and 7 producing identical Macro F1 (std = 0.0);
- post-training abstention, dataset-transparency, annotation-policy, TBD-audit, and S12/S12B comparison reports;
- run metrics, confusion matrix, diagnostics, and reproducibility commands.

This structure is part of the evaluation contribution. It exposes the evidence needed to inspect operating behavior rather than only the final checkpoint.

Artifact type	Review purpose
Metrics reports	Verify full-split accuracy, Macro F1, and per-class behavior
Confusion matrices	Inspect movement among YES, NO, and TBD outcomes
Calibration data	Assess whether confidence supports threshold policies
Threshold sweeps	Select operating points for automation versus deferral
Stability checks	Confirm deterministic evaluation behavior for the release
Reproducibility commands	Reconstruct the evaluated checkpoint and procedure
Post-training tests	Review abstention, dataset, annotation, and error-audit evidence

9.1 Why Audit Evidence Matters

For decision-control models, reproducibility and auditability are not secondary documentation tasks. They determine whether a model can be reviewed before deployment. A headline score does not show whether the confidence distribution is calibrated, whether thresholds were selected after inspecting tradeoffs, whether results are stable across seeds, or whether the exact evaluated artifact can be located.

The evaluation is therefore recorded as a set of review artifacts. The evidence directory is designed to answer reviewer questions: Which checkpoint was evaluated? Which data split was used? What were the per-class outcomes? What threshold sweeps were run? How stable are the results? Where are the reproducibility commands? This makes the reported result more transparent than a standalone benchmark table.

10 Discussion

The results show that bounded decision control can be evaluated as a decision interface rather than as ordinary classification alone. S12B achieves balanced performance across YES, NO, and TBD, with NO F1 = 0.8486 and TBD F1 = 0.7956 on the full test split. The remaining difficulty is the boundary between actionable decisions and deferral, especially ambiguous cases where a human reviewer or downstream policy may prefer conservative routing.

Calibration, threshold sweeps, and multi-seed stability make the model's operating behavior inspectable. Downstream systems can select operating thresholds and document tradeoffs without retraining the core model.

The auxiliary architecture also matters, but empirical claims should remain scoped. In S12B, decision behavior is validated; emotion metrics are not reported because the emotion head is masked. Future releases can separately validate auxiliary value and emotion behavior as independent outputs.

The result is not an isolated leaderboard metric. It is a learned-deferral decision model evaluated with artifacts that describe classwise behavior, calibration, threshold tradeoffs, stability, and artifact identity. These materials are necessary when a model is used as a decision interface rather than a passive classifier.

10.1 Interpretation of the Score Plateau

The S12B result is close to the preceding S12 checkpoint in raw Macro F1, but it is better supported by evaluation evidence because the decision metrics are accompanied by corrected evaluation behavior, calibration evidence, stability checks, abstention baselines, error audits, and a clearer artifact structure. This distinction matters. A slightly higher score without reliable evaluation handling, traceable artifacts, or threshold evidence would be less informative for external review.

The observed plateau near the low-0.82 Macro F1 range suggests that remaining gains are concentrated in the semantic deferral boundary rather than in generic optimization. The model already distinguishes NO well. The hard cases are those where the input contains familiar words or plausible structure but does not supply enough evidence for action. Further improvement is therefore more likely to come from better boundary labeling, domain-specific validation, longer-context handling, and operating-threshold selection than from simply increasing dataset size.

10.2 Implications for Decision Workflows

The practical implication is that AI decision systems should expose uncertainty as a routable outcome. A deployment owner can then decide how uncertain cases move through the workflow: manual review, evidence request, policy escalation, or rejection by default. This is different from treating uncertainty as a hidden property of a model confidence score.

The model makes this separation explicit. It learns a deferral class, while the operating policy selects thresholds and routing behavior. This separation supports review because the model artifact and the deployment rule can be examined independently. If an organization changes its risk tolerance, it can adjust thresholds and document the change without altering the underlying checkpoint.

11 Limitations

Results are specific to DS-L, the S12B checkpoint, and the evaluation setup reported here. Additional domains require separate validation. Inputs are truncated to 128 tokens, so long-context use cases require chunking or preprocessing. The emotion head is masked in this lineage, so this paper does not claim validated emotion-head performance for S12B. The dataset composition includes public NLP sources and curated decision examples, but raw training data is not redistributed in this repository. Users should consult original dataset licenses before reusing source datasets.

The paper also does not claim that TBD is always the correct action under uncertainty. TBD is a model output and an operating-policy option, not a substitute for domain governance. Deployment owners must decide how deferred cases are routed and what thresholds are acceptable for their risk environment. Similarly, the reported calibration value is tied to the validation split and should be remeasured after domain transfer, threshold changes, or additional fine-tuning.

The abstention experiments are diagnostic baselines, not a complete comparison against independently trained abstention systems. The forced binary baseline tests the loss of the TBD class, while confidence, entropy, and margin rejection test retained-set behavior under thresholded abstention. A stronger conference version should include separately trained binary and selective-classification baselines, conformal abstention, and domain-transfer evaluations.

Finally, the audit evidence should be understood as a review aid, not as a guarantee of universal safety. It makes the evidence inspectable, but each deployment still requires application-specific validation and monitoring.

12 Conclusion

This paper presented a bounded decision-control model with learned deferral. The motivating problem is that operational AI systems often need to decide whether to act, reject, or defer, and a forced binary or generative output can hide uncertainty from the surrounding workflow. The model addresses this problem with a three-way learned decision interface: YES, NO, and TBD.

On the full DS-L test split, S12B achieves Accuracy = 0.8260 and Macro F1 = 0.8252, with per-class F1 scores of 0.8314 (YES), 0.8486 (NO), and 0.7956 (TBD). A forced binary YES/NO view of the same task drops to Macro F1 = 0.4945 because it cannot represent deferral. The S12B boundary-pack pass does not materially improve raw F1 over S12, but it reduces calibration error and high-confidence error rates. The result is accompanied by calibration evidence, threshold sweeps, stability checks, confusion matrices, abstention baselines, error audits, and reproducibility materials.

The paper's claim is therefore not only that the model reaches a particular score, but that learned deferral can be evaluated as an auditable decision-control surface.

The remaining work is clear: improve the semantic boundary between direct action and deferral, validate auxiliary value and emotion outputs as independent empirical claims, and repeat calibration and threshold review for each deployment domain. Within the evaluated scope, S12B provides a concrete checkpoint for bounded decision routing under ambiguity.

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